

10 Three planes have equations

$$-x + 2y + z = 0$$

$$2x - y - z = 0$$

$$x + y = a$$

where a is a constant.

(i) Investigate the arrangement of the planes:

- when $a = 0$;
- when $a \neq 0$.

[6]

(ii) Chris claims that the position vectors $-\mathbf{i} + 2\mathbf{j} + \mathbf{k}$, $2\mathbf{i} - \mathbf{j} - \mathbf{k}$ and $\mathbf{i} + \mathbf{j}$ lie in a plane. Determine whether or not Chris is correct.

[2]